

PhD Research Proposal

**Transdiagnostic Dimensions of Adolescent Psychopathology: Cognitive, Longitudinal, and
Neural Characterisation**

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Introduction

Mental health disorders are predominantly developmental: half of all lifetime conditions emerge before age 14, and three-quarters by age 24. Yet mechanistic understanding of adolescent psychopathology remains poor, and existing treatments produce only modest effects. A core obstacle is that current diagnostic frameworks impose categorical boundaries that do not reflect the dimensional, transdiagnostic nature of adolescent symptoms.

Computational Factor Modelling (CFM; Gillan et al., 2016) offers a principled solution. By applying factor analysis to symptom covariance and arbitrating the resulting dimensions against computationally derived cognitive parameters, CFM identifies not only that transdiagnostic dimensions exist, but which cognitive mechanisms characterise them. The approach has been established across multiple adult populations, with two core dimensions identified: Compulsive Behaviour and Intrusive Thought, anchored to reduced model-based planning, and Anxious-Depression, anchored to metacognitive underconfidence (Wise et al., 2023).

CFM has never been applied to adolescents. Whether its dimensions exist in younger populations, and whether their cognitive signatures carry the same mechanistic meaning during a period of active brain development, is entirely unknown. This proposal addresses that gap through three objectives. First, to identify transdiagnostic CFM dimensions in a large UK adolescent cohort and examine their cognitive grounding. Second, to determine whether those dimensions are stable or dynamic across adolescent development using a longitudinal design. Third, to examine whether the dimensions have a neural substrate, using Precision Functional Mapping to characterise individual-level functional network organisation and its association with dimension scores. Establishing that CFM dimensions exist, are stable, and are neurally grounded

in adolescents would provide a mechanistically anchored account of transdiagnostic psychopathology during this developmental period, identifying which cognitive processes drive symptom organisation at the stage when intervention is most likely to alter trajectory.

Literature Review

Adolescence

Adolescence is a period of unique neurobiological and psychosocial vulnerability. Half of all lifetime mental health conditions emerge before age 14, and three-quarters by age 24 (Kessler et al., 2005). Yet mental health needs in this group remain profoundly unmet: the median delay between symptom onset and treatment is 11 years (Colizzi et al., 2020; Kessler et al., 2005). Additionally, current treatments produce only modest effects, with even poorer outcomes when delivered outside controlled research settings (Weisz et al., 2017). Improving treatment outcomes requires identifying the mechanisms that drive symptom development (Holmes et al., 2018), but the characteristic fluidity of adolescent psychopathology makes it exceptionally difficult. High comorbidity, diagnostic instability, and heterotypic continuity mean that categorical frameworks impose diagnostic boundaries that do not capture how symptoms actually co-occur in this population (Caspi & Moffitt, 2018; Copeland et al., 2013; Shah et al., 2020).

Transdiagnostic Psychiatry

The field has responded with two influential dimensional frameworks. The Research Domain Criteria (RDoC)(Insel et al., 2010) organises psychopathology around neurobiological constructs: motivation, cognition, social processes that cut across diagnostic categories. The Hierarchical Taxonomy of Psychopathology (HiTOP)(Kotov et al., 2017) organises symptoms

empirically, using factor analysis to derive a hierarchy of dimensions from patterns of symptom co-occurrence. Both have generated substantial empirical support, but a systematic review found that 81% of transdiagnostic studies remain purely descriptive and only 7% reach the level of causal inference (Fusar-Poli et al., 2019). The dimensions exist and replicate, but they are not mechanistically explained. A related debate concerns whether psychopathology is best captured by a single general liability factor: the p-factor, or by specific, separable dimensions that carry independent mechanistic meaning (Caspi & Moffitt, 2018); resolving this requires arbitrating dimensional solutions against independent cognitive or neural evidence. A descriptive taxonomy, however well-characterised, cannot specify which cognitive processes to target in treatment, nor can it distinguish individuals who share similar symptom profiles for different underlying reasons. This mechanistic gap remains one of the central unresolved problems in transdiagnostic psychiatry.

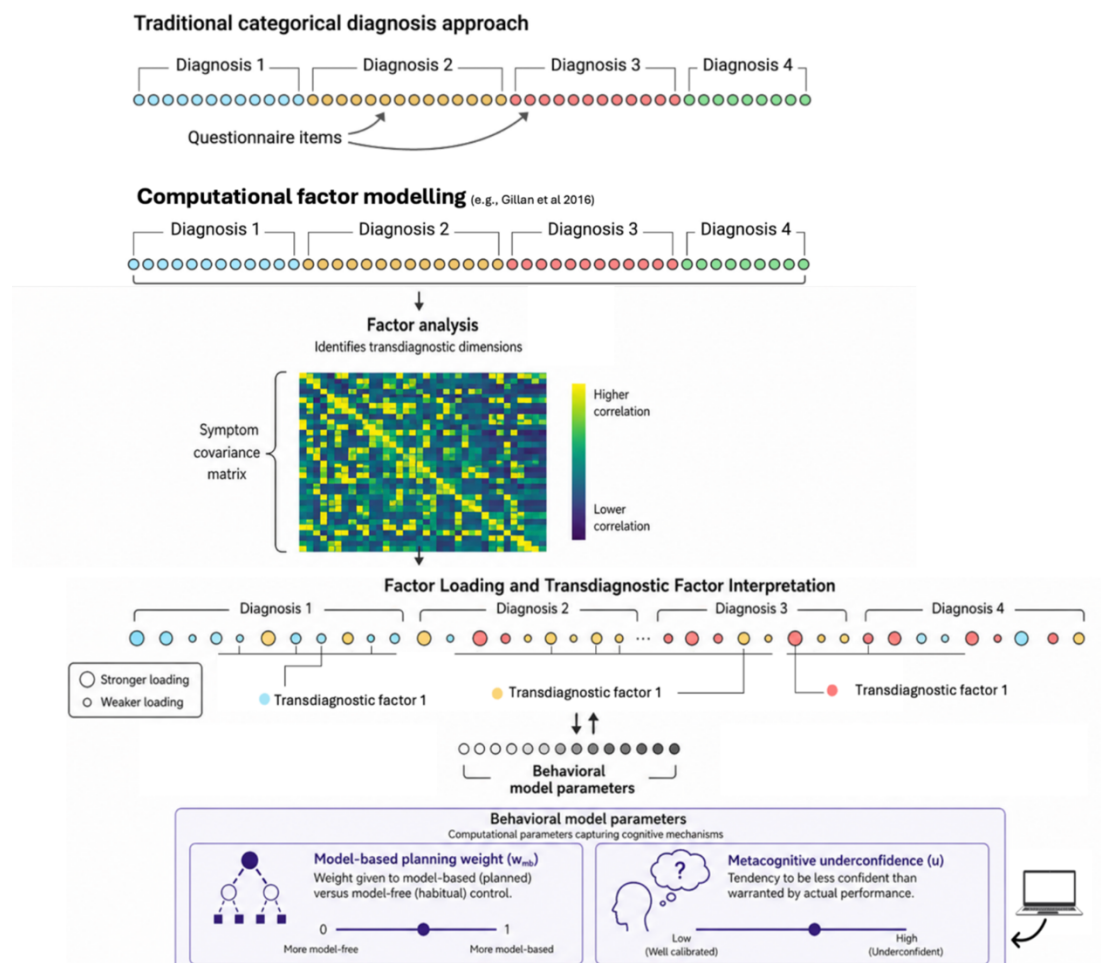
Computational Factor Modelling

Computational Factor Modelling (CFM; Figure 1) was developed to address this gap. Introduced by Gillan et al. (2016), CFM applies factor analysis to a symptom covariance matrix pooled across diagnostic categories, deriving transdiagnostic dimensions from questionnaire data spanning multiple disorder domains. Critically, the resulting dimensions are arbitrated against computationally derived cognitive parameters, which represent quantitative estimates of specific psychological processes extracted from behavioural tasks and formal computational models. This arbitration step is what distinguishes CFM from RDoC and HiTOP: rather than inferring mechanism from symptom structure, it tests mechanistic hypotheses directly against behavioural parameters. The computational psychiatry framework underlying this approach argues that

formalising cognitive mechanisms mathematically is a necessary step toward developing targeted interventions (Huys et al., 2016).

Figure 1

Computational factor modelling (CFM) pipeline.



Note. Adapted from Wise et al., (2023). CFM pools questionnaire items across diagnostic categories and applies factor analysis to the symptom covariance matrix to identify transdiagnostic dimensions. Dimensions are then arbitrated against behavioural model

parameters, such as the model-based planning weight and metacognitive underconfidence, grounding each dimension in a specific cognitive mechanism.

In the original study (N=1,413 adults), three transdiagnostic dimensions emerged: Compulsive Behaviour and Intrusive Thought (CIT), Anxious-Depression (AD), and Social Withdrawal. CIT was associated with reduced model-based planning, the capacity to use an internal model of the environment to guide decision-making, assessed via a two-step reinforcement learning task (Gillan et al., 2016). AD was subsequently anchored to metacognitive underconfidence: a tendency to underestimate the reliability of one's own perceptual judgements, assessed via a roulette task (Wise et al., 2023). Notably, Social Withdrawal did not yield a robust cognitive anchor, illustrating an important limitation: not all transdiagnostic dimensions may be amenable to computational arbitration in their current form. A multiverse analysis confirmed that the CIT and AD findings are robust: CFM dimensions outperformed alternative analytical choices across 99.88% of questionnaire combinations tested, with questionnaire selection, not cognitive task choice, emerging as the most consequential analytical decision (Fox et al., 2026).

To date, however, CFM has been applied exclusively to adult samples. More broadly, while descriptive dimensional approaches have characterised transdiagnostic symptom structures in younger populations (Carragher et al., 2016), mechanistic characterisation has not followed. The specific step of grounding transdiagnostic dimensions in computationally derived cognitive parameters has not been extended to adolescent populations. Whether adolescent psychopathology organises along similar transdiagnostic lines, and whether those dimensions can be anchored to the same cognitive mechanisms identified in adults, remains entirely unknown. It is not safe to assume adult findings generalize. First, symptom covariance structures

themselves may differ developmentally, as several symptom domains associated with adult psychopathology dimensions emerge or consolidate across adolescence. Second, the cognitive processes that arbitrate adult dimensions are developmentally regulated. Prefrontal systems supporting model-based planning mature later than limbic and striatal systems involved in reward processing, contributing to characteristic shifts in adolescent exploratory and goal-directed behaviour (Mills et al., 2014; Paus et al., 2008). Importantly, adolescents show heightened exploratory behaviour that, when assessed using the same model-based planning parameter used in adult work, resembles impaired goal-directed control, but in the adolescent context, this may reflect a normative developmental profile rather than a marker of pathology (Lloyd et al., 2021). This matters for interpretation: an adolescent who scores high on a dimension resembling adult compulsivity may do so for a different reason than an adult with the same profile. Applying adult-derived intervention targets without establishing this would risk clinical misapplication.

Developmental Trajectory

The broader developmental psychopathology literature shows that transdiagnostic dimensions are not static across adolescence: longitudinal work identifies distinct trajectory subgroups, including increasing, persisting, and declining courses, within the same broad dimension across childhood and adolescence (Reef et al., 2011). Consistent with this, the clinical staging model formalises the prognostic significance of developmental trajectory: subthreshold presentations may resolve, persist, or progress across adolescence, and the trajectory, not the cross-sectional observation, is what carries clinical meaning (Shah et al., 2020). Yet knowing that developmental trajectories carry prognostic significance is not the same as understanding how computationally grounded transdiagnostic dimensions themselves evolve across adolescence.

CFM dimensions carry a layer of complexity that general symptom trajectories do not: not only may the symptom factor structure shift across adolescence, but the cognitive mechanisms that anchor each dimension are themselves developmentally regulated. Whether a computationally grounded dimension score reflects the same mechanistic process at age 14 as it does at age 24, or whether both the dimension and its cognitive anchor change across this period, is a question that only longitudinal data can address. The CFM literature has relied on cross-sectional designs in adult samples (Wise et al., 2023) and has not been applied to any developmental population, leaving the longitudinal behaviour of computationally grounded dimensions in adolescents uncharacterised and motivating the longitudinal design of the present project.

Neural Basis of Transdiagnostic Dimensions

The neural basis of adolescent transdiagnostic dimensions represents a third and complementary gap. Although dimensional frameworks have been linked to behavioural manifestations, less is known about how they relate to individual differences in brain organisation across development. Current evidence raises several unresolved questions. Neural-psychopathology associations appear developmentally sensitive: the magnitude of brain-behaviour relationships vary across developmental stages (Satterthwaite et al., 2013), and their direction is not uniform across age groups, with the same neural feature carrying different clinical implications in adolescent versus adult samples (Vanes & Dolan, 2021). Most directly, a systematic review concluded that neural correlates of transdiagnostic dimensions are sparse and inconsistent, with no longitudinal evidence linking dimensional structure to changes in brain organisation over time (Hoy et al., 2023; Vanes & Dolan, 2021). These null and inconsistent findings may reflect, in part, a fundamental methodological constraint: conventional group-level neuroimaging averages over participants, implicitly assuming that the neural substrates of

transdiagnostic dimensions are uniform across individuals. Converging evidence challenges this assumption. Large-scale functional network architecture exhibits stable inter-individual variability, including differences in network boundaries and regional network assignment that are reproducible within individuals over time (Dworetsky et al., 2024; Seitzman et al., 2019). Such variability may be particularly relevant in developmental populations, where ongoing neurodevelopmental change could alter how psychopathology dimensions are instantiated neurally across individuals. Although conventional group-level neuroimaging approaches can identify population-average associations, they may be less sensitive to fine-grained individual variation in functional topology that could carry mechanistically informative signal.

Precision Functional Mapping

Precision Functional Mapping (PFM) addresses this challenge by characterising each individual's functional network architecture directly, rather than averaging across participants. Individual-level network variants take two principal forms: border shifts, in which network boundaries are displaced from population-average locations, and ectopic intrusions, in which regions associated with one network appear within the territory of another (Dworetsky et al., 2024). In adult samples, these variants have been shown to be stable within individuals over time, suggesting they are trait-like features of brain organisation, though whether this stability holds in adolescent populations remains to be established. PFM is particularly well-suited to adolescent samples, whose functional networks are undergoing substantial maturation and increasing specialisation, rendering group-averaged connectivity estimates especially liable to conflate developmental change with individual differences (Mills et al., 2014). PFM therefore provides a framework for testing whether CFM dimensions are associated with distinct and

stable patterns of individual-level network organisation, a question that group-averaged connectivity approaches are structurally unable to address.

Rationale and Aims

Three gaps converge to define the present proposal. CFM has never been applied to adolescents, leaving the structure and cognitive grounding of transdiagnostic dimensions in this developmental period entirely unknown. Even if such dimensions can be identified, their longitudinal dynamics, whether they are stable or shifting across adolescence, have not been characterised. And the individual-level neural architecture associated with adolescent dimensions, including whether neural organisation precedes or follows symptomatic change, remains unexamined.

Resolving these gaps would produce the first mechanistically grounded characterisation of transdiagnostic psychopathology in adolescence, a foundation for identifying which cognitive processes to target during the developmental window when intervention is most likely to prevent progression from subthreshold distress to clinical disorder. It would also establish whether the computational arbitration approach that anchors CFM dimensions to specific cognitive mechanisms in adults can be meaningfully extended to a younger population, a question with direct consequences for whether mechanistically targeted interventions are, in principle, achievable in adolescent psychiatry.

Proposed Method

Overview

This project employs a quantitative, secondary analysis design, drawing on pre-existing cohort data from the Neuroscience in Psychiatry Network (NSPN). The NSPN design combines

cross-sectional age variation with repeated within-person assessments approximately 18 months apart, enabling the study of developmental change across adolescence and early adulthood without requiring decade-long follow-up periods. This design is particularly well suited to the present project, which aims to examine the emergence, stability, and neural correlates of computationally grounded transdiagnostic dimensions across development. The project is organised across three linked aims: (1) identifying computationally grounded transdiagnostic dimensions in adolescence (Figure 2A), (2) examining their longitudinal developmental stability (Figure 2B), and (3) characterising their neural correlates using individualized functional network mapping approaches (Figure 2C). Across all aims, analyses will be conducted using pre-registered analytic pipelines.

Dataset

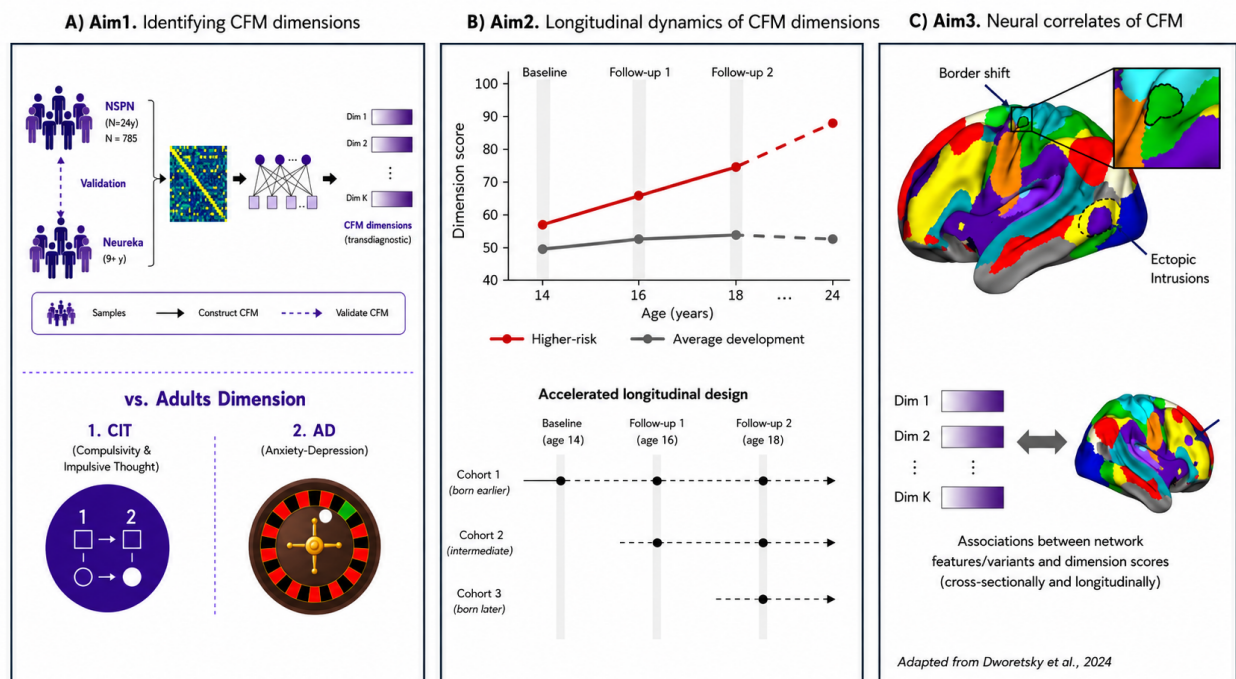
This project draws on three linked cohorts from the Neuroscience in Psychiatry Network (NSPN). The NSPN dataset (Kiddle et al., 2018) comprises a general population sample of 2,402 participants aged 14-24 recruited across three UK sites using an accelerated longitudinal design, with three follow-up waves approximately 18 months apart. Two nested subsamples provide the primary data for the present project.

The NSPN Cognition Cohort (N=785 baseline, N=568 follow-up; 72% retention) completed a comprehensive cognitive task battery. Questionnaires cover symptom domains including obsessive-compulsive symptoms, depression, anxiety, impulsivity, schizotypy, and alcohol use. The cognitive battery includes tasks assessing model-based planning, metacognition, inhibitory control, approach-avoidance behaviour, information gathering, temporal discounting, and social decision-making. The co-location of questionnaire and cognitive data in the same sample enables both symptom factor analysis and cognitive arbitration within a single cohort.

The NSPN MRI Cohort (N=318 baseline, N=234 follow-up; 74% retention; median interval 15.4 months) is a nested subsample of the Cognition Cohort that additionally completed multi-echo resting-state functional MRI. Multi-echo acquisition substantially reduces thermal noise and susceptibility artefacts, making it particularly well-suited for individualised functional connectivity and network topology analyses.

Figure 2

Schematic of Proposed Project



Aim 1 Identifying transdiagnostic dimensions of psychopathology in adolescents

Aim 1 asks whether cognitively meaningful transdiagnostic dimensions can be identified during adolescence and early adulthood. To address this question, the CFM pipeline will be applied to wave 1 of the NSPN Cognition Cohort (N=785), integrating dimensional symptom questionnaires with computational parameters derived from cognitive task performance,

following the approach of Gillan et al. (2016). This approach allows symptom covariance structure to be evaluated alongside cognitive processes implicated in computational psychiatry, enabling identification of dimensions grounded not only in symptom co-occurrence but also in cognitive mechanisms. If data supports, more advanced analysis would be conducted following Fox et al. (2026): a multiverse approach will be implemented across varying questionnaire combinations to identify if the dimension solution determined is most robust across analytical choices.

Measurement invariance testing will then assess whether identified dimensions carry comparable meaning across the 14–24 developmental range. Configural and metric invariance will be tested using confirmatory factor analysis to determine whether symptom items load onto dimensions similarly across developmental stages, ensuring that any comparisons of dimension scores across age groups are statistically interpretable.

The resulting adolescent CFM dimensions will be characterised by the symptom domains they capture, noting where these correspond to or diverge from the three adult dimensions identified by Gillan et al. (2016): anxious-depression, compulsive behaviour and intrusive thought, and social withdrawal. This provides an initial indication of whether similar broad symptom domains emerge in adolescents, with formal structural comparison reserved as a future direction given difference in questionnaire batteries across the two studies.

Aim 2: Longitudinal dynamics of adolescent CFM dimensions

Aim 2 examines whether CFM dimensions identified in Aim 1 are stable or dynamic over time. Using the NSPN Cognition Cohort, intraclass correlations will quantify how much

variance in dimension scores is stable across waves versus occasion-specific, testing whether high scores reliably identify the same individuals over time.

Secondly, cross-lagged panel models will test whether baseline dimension scores are associated with wave 2 and 3 cognitive performance controlling for baseline performance. Normative developmental trajectories will be accounted for by residualizing cognitive change against expected growth.

Dimension-cognition associations may not be uniform across the full adolescent period. Age will be entered as a continuous moderator to test whether these associations differ across the adolescent period, with sensitivity analyses comparing younger (14–18) and older (19–24) subgroups.

Aim 3: Neural correlates of adolescent CFM dimensions

Aim 3 asks whether the CFM dimensions identified in adolescence map onto distinct patterns of functional network organisation, and whether these neural correlates are stronger and more dimensionally specific than associations observed using symptom questionnaires in their original DSM-aligned form.

Resting-state fMRI data from the NSPN MRI Cohort (N = 318 baseline) will be analysed using PFM (Seitzman et al., 2019) to test whether different dimensions are associated with distinct patterns of individual-level network organisation, characterised by border shifts and ectopic intrusions (Dworetsky et al., 2024), and whether individual-level network organisation reflects dimension scores. Comparative analyses will examine whether CFM-derived dimension scores explain more variance in individual-level network organisation than raw symptom questionnaire scores, testing whether computational dimensionality reduction yields stronger and

more dimensionally specific neural associations. Models including CFM dimensions will be compared with models including original questionnaire and domain scores using out-of-sample predictive performance and effect size comparisons, with nested model tests used only where predictors are hierarchically specified within the same model.

Longitudinal analyses, using waves 2 and 3 of the NSPN MRI cohort, will address two further questions. First, whether individual-level functional topology remains stable within individuals across development (median 15-month inter-wave interval). Second, whether baseline CFM dimension score is prospectively associated with change in network organisation by wave 2, testing whether transdiagnostic dimensions carry prognostic signal for neural developmental trajectories. Given the modest sample size for individual-level neuroimaging analyses, Aim 3 findings will be interpreted in light of expected effect sizes in the brain-behaviour literature.

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